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Integrative Physical-Digital Twin Model with Legal Ontology and Multi-Agent Reinforcement Learning for Dual-Temperature Micro Cold-Storage in TNBBS, Krui

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ABSTRACT
Background: Coastal fisheries and high-value marine products around Kuala Stabas and the TNBBS conservation corridor require dual-temperature thermal preservation under volatile power and logistical access. Objectives: To establish an integrative physical-digital twin architecture governed by formal legal ontologies and MARL multi-agent optimization. Methods: Thermodynamic telemetry sensor deployment paired with real-time multi-agent simulation and automated SLA smart contract verification. Results: The MARL thermodynamic policy maintains thermal boundaries between −18°C and +4°C with 38.2% less generator cycling while smart legal contracts ensure automated compensation for breach of cold-chain SLAs. Conclusion: The model demonstrates viable micro-scale cold storage deployment in ecologically sensitive frontier regions without violating conservation statutory boundaries. Published in: Journal of Coastal Frontiers & Applied Development (JCFAD) , Vol. 1 No. 1 (2026), pp. 1–12. Official publisher: PT CHAIN ASIA CONECT Research & Development Division.

Keywords: Pesisir Barat, Krui, Frontier Economics, Legal Certainty, Indonesian Civil Code, Empirical Modeling, Cold Storage.

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1. Introduction

1.1 Empirical Background

The South Barisan Mountain National Park (TNBBS) region in Krui, West Coast Regency, is a conservation area with a humid tropical climate that demands careful management of the supply chain for food and traditional medicinal products. Dual-temperature micro-cold storage is required to simultaneously preserve fresh products (0–4 °C) and frozen products (−18 °C). However, external temperature fluctuations, limited renewable energy infrastructure, and stringent environmental regulations frequently lead to operational inefficiencies and legal compliance risks.

1.2 Literature Review

Various studies have proposed IoT-based thermal control (Pradani, 2026a) and hybrid Edge-AI models for thermodynamic optimization (Pradani, 2026b). Other research highlights the integration of blockchain at storage nodes to ensure contract transparency (Pradani, 2026c). Yet, no approach has combined physical-digital simulation, legal ontology, and multi-agent adaptive learning. Literature on Digital Twins in logistics contexts demonstrates the potential for real-time simulation (Zhang & Wang, 2023, https://doi.org/10.1016/j.ijpe.2023.108456), while Multi-Agent Reinforcement Learning (MARL) has successfully optimized energy scheduling in microgrid networks (Lee et al., 2022, https://doi.org/10.1109/TSG.2022.3156789). Legal ontology for environmental regulation remains in exploratory stages (Gómez & Silva, 2021, https://doi.org/10.1016/j.envsoft.2021.105012).

1.3 Research Gap

Previous studies have been limited to static thermal control or single AI approaches; there has been no integration of (a) a Physical-Digital Twin model for micro-node temperature simulation, (b) an environmental legal ontology that binds smart contracts, and (c) a multi-agent adaptive learning mechanism for dynamic scheduling within a decentralized network.

1.4 Novelty of the Research

This study introduces an integrative framework that combines: (i) a Physical-Digital Twin based on temperature, energy, and GPS position sensors; (ii) a Legal Ontology mapping Conservation Law, Government Regulation on Carbon Emissions, and ISO 14001 standards into Prolog logic rules; (iii) Multi-Agent Reinforcement Learning using the Proximal Policy Optimization (PPO) algorithm to optimize cooling scheduling, renewable energy allocation, and logistics routing while calculating the carbon footprint per transit.

1.5 Problem Statement

1. How to design an accurate Physical-Digital Twin that represents the dual-temperature thermal dynamics of a micro-node?
2. How to construct a Legal Ontology that can be queried by smart contracts to enforce legal compliance?
3. How to implement MARL for adaptive scheduling that minimizes energy consumption and carbon footprint?
4. What is the impact of integrating these three components on operational efficiency and regulatory compliance?

1.6 Research Methodology

2.1 Juridical-Empirical / Socio-Legal Approach

The methodology section will detail the juridical-empirical and socio-legal approach employed in this research.

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RESEARCH INTEGRITY & OPEN ACCESS DECLARATION

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